Uploading Files to Rails Applications

Your application may allow users to upload files. For example, a bug-reporting system might let users attach log files and code samples to a problem ticket, or a blogging application could let its users upload a small image to appear next to their articles.

In HTTP, files are uploaded as a *multipart/form-data* `POST` message. As the name suggests, forms are used to generate this type of message. Within that form, you'll use `<input>` tags with `type="file"`. When rendered by a browser, this allows the user to select a file by name. When the form is subsequently submitted, the file or files will be sent back along with the rest of the form data.

To illustrate the file upload process, we'll show some code that allows a user to upload an image and display that image alongside a comment. To do this, we first need a `pictures` table to store the data:

rails80/e1/views/db/migrate/20170425000004_create_pictures.rb

```
class CreatePictures < ActiveRecord::Migration
  def change
    create_table :pictures do |t|
      t.string :comment
      t.string :name
      t.string :content_type
      # If using MySQL, blobs default to 64k, so we have to give
      # an explicit size to extend them
      t.binary :data, :limit => 1.megabyte
    end
  end
end
```

We'll create a somewhat artificial upload controller just to demonstrate the process. The `get` action is pretty conventional; it simply creates a new picture object and renders a form: